

Existing tests cover specific system guarantees



EVALUATION · TEST CASE, IMPLEMENTATION AND TEST LEVEL

Representative cases link each requirement to an implemented check.

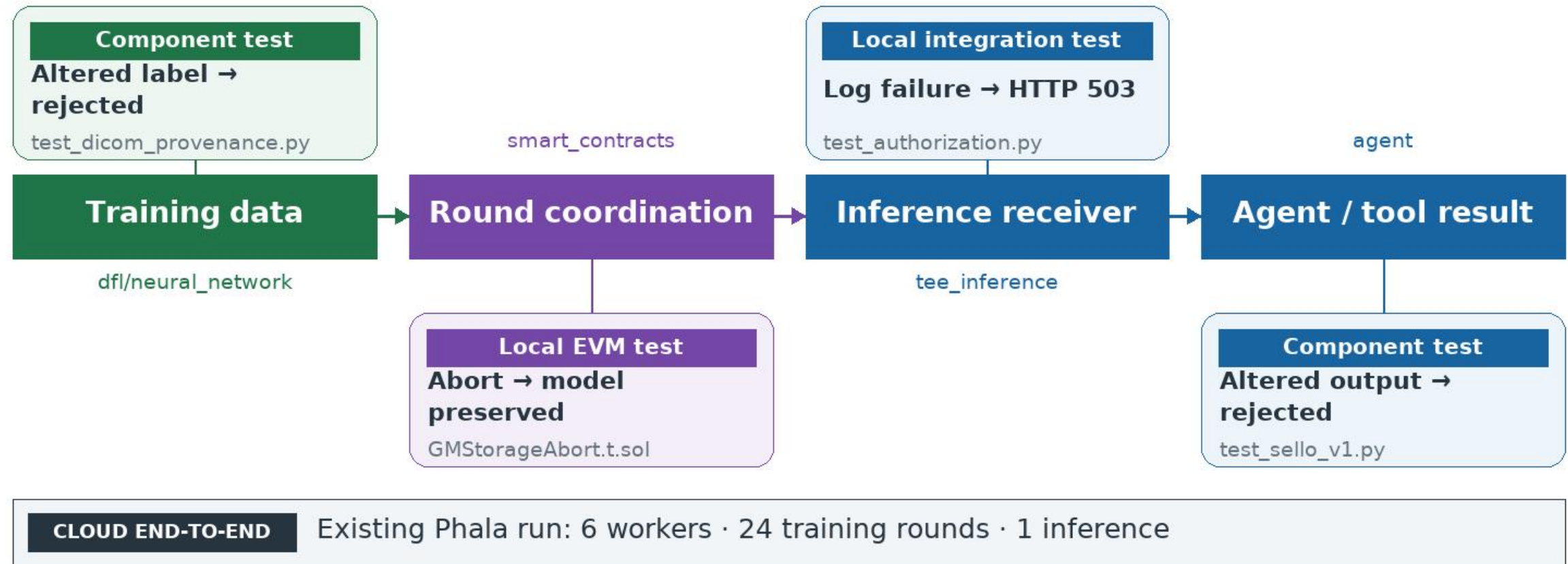
BOUNDARY	EXAMPLE ASSERTION	IMPLEMENTED IN	TEST TYPE
Training input	Changed signed label is rejected	dfi/neural_network/tests/ test_dicom_provenance.py	Component Python · signer fixtures
Round state	Abort preserves the last finalized model	smart_contracts/test/ GMStorageAbort.t.sol	Contract component Foundry · local EVM
Model exchange	No recipient key → no decryption	dfi/node_server/test/ gm_crypto.test.js	Component Node.js · local AES/RSA
Inference	Substituted request is rejected	agent/tests/ test_tee_inference_client.py	Component Python · synthetic quote
Receiver API	Log failure → HTTP 503 and no receipt	tee_inference/tests/ test_authorization.py	Local integration ASGI · mocked log

Separate end-to-end evidence: one Phala run · 24 training rounds · final inference

Scope: local fixtures and mocked dependencies; live aggregator failover was not exercised.

Evaluation along the architecture

IMPLEMENTED CHECKS AT FOUR SYSTEM BOUNDARIES



Local test cases use controlled dependencies. The cloud run demonstrates the deployed workflow; recovery was not triggered.

What happens when evidence is wrong?

EXISTING TESTS · THREE CONTROLLED FAULT SCENARIOS

01 TRAINING DATA

- **Modified signed label**

test_dicom_provenance.py

COMPONENT UNDER TEST

Training provenance verifier

Component · Python

EXPECTED RESPONSE

Reject invalid signature

02 INFERENCE EVIDENCE

- **Substituted inference request**

test_tee_inference_client.py

COMPONENT UNDER TEST

AIR bundle verifier

Component · Python

EXPECTED RESPONSE

Reject request mismatch

03 TOOL COMPLETION

- **Log publication failure**

test_authorization.py

COMPONENT UNDER TEST

Receiver API

Local integration · ASGI, mocked log

EXPECTED RESPONSE

**HTTP 503
No receipt**

Implemented scenarios; expected outcomes shown. Controlled dependencies; no new test-run results.

Different tests answer different questions

THREE COMPLEMENTARY LEVELS OF EVIDENCE

Checks explain individual decisions.
The deployed run demonstrates composition.

01 COMPONENT / CONTRACT

**Do the checks
reject invalid state?**

Modified input rejected;
last finalized model preserved.

`.../tests/test_dicom_provenance.py`
`.../test/GMStorageAbort.t.sol`

02 LOCAL SERVICE

**Does the receiver
fail closed?**

ASGI request with an injected
log-publication failure:

HTTP 503 · no receipt

`tee_inference/tests/`
`test_authorization.py`

03 DEPLOYED WORKFLOW

**Do the components
work together?**

Existing Phala run

6 workers
24 training rounds
1 final inference

No injected failure.

`data/evaluation/`
Phala run record · 2026-09-01

Complementary scopes: controlled checks and local integration support the single deployed functional run.